

Git Hands-On Lab 4 – Resolving Merge Conflicts

Objective

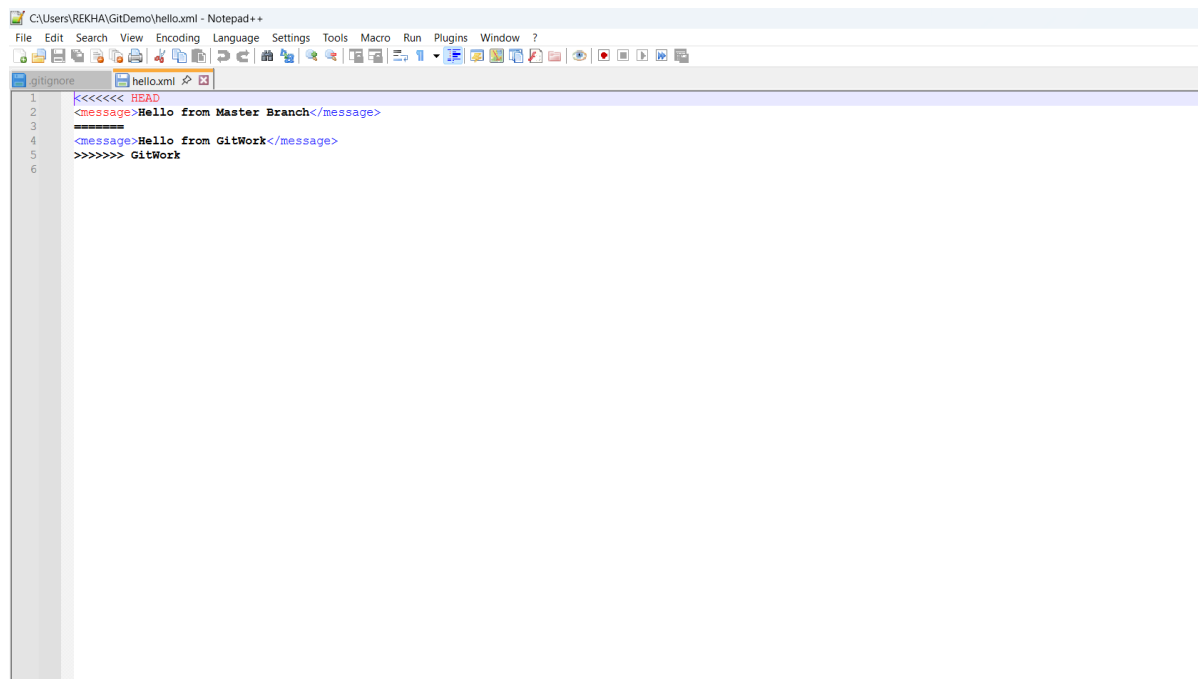
To create a merge conflict between two branches, resolve it manually, update the .gitignore file, and verify the repository status.

Step 1: Create GitWork Branch and Commit

```
cd ~/GitDemo
git status
git branch GitWork
git checkout GitWork
echo "<message>Hello from GitWork</message>" > hello.xml
cat hello.xml
git status
git add hello.xml
git commit -m "Added hello.xml in GitWork"
```

Observed Output:

hello.xml created and committed successfully in GitWork branch.



Step 2: Create Different hello.xml in Master

```
git checkout master
echo "<message>Hello from Master Branch</message>" > hello.xml
cat hello.xml
git status
```

```
git add hello.xml
git commit -m "Added hello.xml in masters"
git log --oneline --graph --decorate --all
git diff master GitWork
```

Observed Output:

The differences between the master branch and GitWork branch are displayed.

```
MINGW64/c/Users/REKHA/GitDemo
REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (GitWork)
$ git checkout master
Switched to branch 'master'

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (master)
$ echo "<message>Hello from Master Branch</message>" > hello.xml

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (master)
$ cat hello.xml
<message>Hello from Master Branch</message>

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (master)
$ git status
On branch master
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    hello.xml

nothing added to commit but untracked files present (use "git add" to track)

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (master)
$ git add hello.xml
warning: in the working copy of 'hello.xml', LF will be replaced by CRLF the next time Git touches it

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (master)
$ git commit -m "Added hello.xml in masters"
[master 0d8fc09] Added hello.xml in masters
1 file changed, 1 insertion(+)
create mode 100644 hello.xml

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (master)
$ git log --oneline --graph --decorate --all
* 0d8fc09 (HEAD -> master) Added hello.xml in masters
| * dce6863 (GitWork) Added hello.xml in GitWork
|/
* 6e823d7 Added branch.txt in GitNewBranch
* 047e05e Added .gitignore file
* 190ed62 Initial commit

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (master)
$ git diff master GitWork
diff --git a/hello.xml b/hello.xml
index e3bbec4..84f3217 100644
--- a/hello.xml
+++ b/hello.xml
```

Step 3: Merge Conflict

```
git merge GitWork
```

Observed Output:

CONFLICT (add/add): Merge conflict in hello.xml

Automatic merge failed; fix conflicts and then commit the result.

```
notepad++ hello.xml
```

```
MINGW64/c/Users/REKHA/GitDemo

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (master)
$ git merge GitWork
Auto-merging hello.xml
CONFLICT (add/add): Merge conflict in hello.xml
Automatic merge failed; fix conflicts and then commit the result.

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (master|MERCING)
$ notepad++ hello.xml

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (master|MERCING)
$ git add hello.xml

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (master|MERCING)
$ git commit -m "Resolved merge conflict"
[master fddd191] Resolved merge conflict

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (master)
$ notepad++ .gitignore

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (master)
$ git add .gitignore

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (master)
$ git commit -m "Updated .gitignore"
[master 85da0ee] Updated .gitignore
1 file changed, 2 insertions(+), 1 deletion(-)

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (master)
$ git branch
  GitWork
* master

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (master)
$ git branch -d GitWork
Deleted branch GitWork (was dce6863).

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (master)
$ git status
On branch master
nothing to commit, working tree clean

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (master)
$ git log --oneline --graph --decorate
* 85da0ee (HEAD -> master) Updated .gitignore
* fddd191 Resolved merge conflict
```

Step 4: Resolve Conflict

Edit hello.xml and replace the conflict markers with:

```
<message>Hello from Master Branch</message>
<message>Hello from GitWork</message>
```

```
git add hello.xml
git commit -m "Resolved merge conflict"
```

```
C:\Users\REKHA\GitDemo\hello.xml - Notepad++
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?
[Icons]
gitignore hello.xml x
1 | <message>Hello from Master Branch</message>
2 | <message>Hello from GitWork</message>
```

Step 5: Update .gitignore and Verify

```
notepad++ .gitignore
```

Add:

```
*.orig
```

```
git add .gitignore
```

```
git commit -m "Updated .gitignore"
```

```
git branch
```

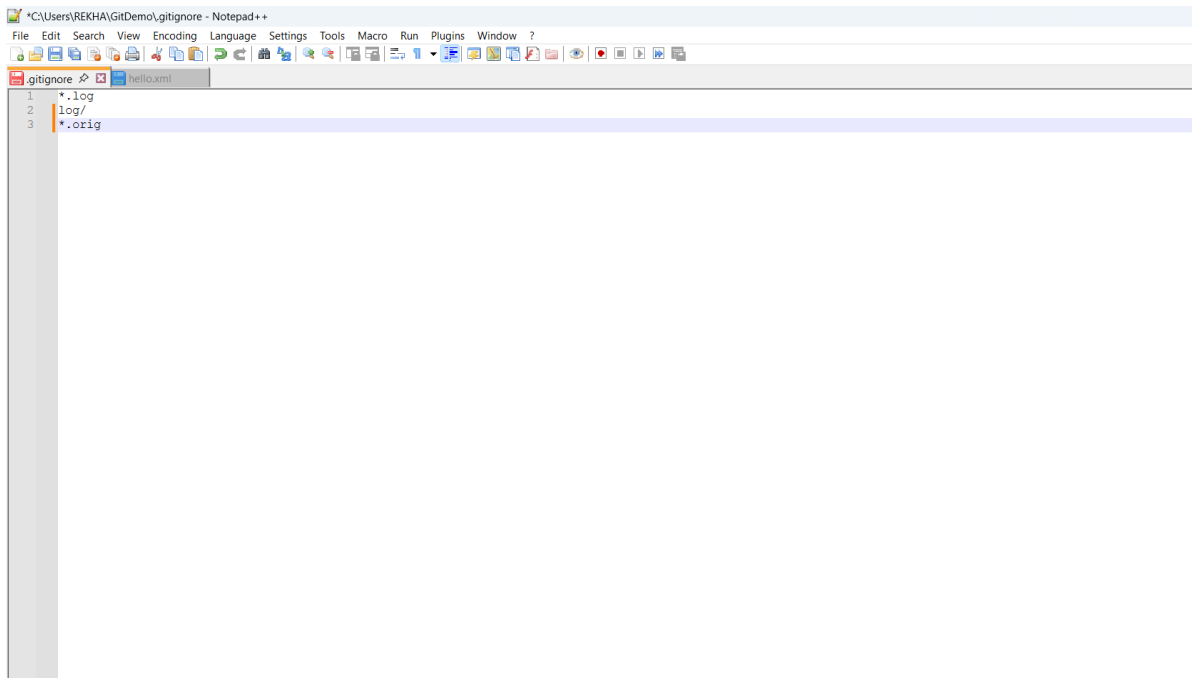
```
git branch -d GitWork
```

```
git status
```

```
git log --oneline --graph --decorate
```

Observed Output:

Repository is clean and GitWork branch is deleted successfully.

A screenshot of the Notepad++ application window. The title bar reads '*C:\Users\REKHA\GitDemo\.gitignore - Notepad++'. The menu bar includes File, Edit, Search, View, Encoding, Language, Settings, Tools, Macro, Run, Plugins, Window, and ?. The toolbar contains various icons for file operations and editing. The editor area shows the content of the .gitignore file with line numbers 1, 2, and 3 on the left margin. The text in the editor is: 1 *.log, 2 log/, and 3 *.orig. The file name '.gitignore' is visible in the tab at the top of the editor.

Result

The merge conflict was resolved successfully, .gitignore was updated, the GitWork branch was deleted, and the repository was verified to be in a clean state.